Erum Hassan

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EDUCATION

University of Oklahoma **Aug 2021 - May 2025**

BS Chemical Engineering (Sustainability), and Data Science and Analytics (GPA: 3.69) Norman, Oklahoma

- **Achievements:** Senior Researcher of the Year (2025), Student Staff Member of the Year (Aug 2023- May 2024), Full Scholarship to the University of Oklahoma, Phillips 66 Research Scholarship, OU Dean's List, Mote Endowed Engineering Scholarship, Dorothy Grace Barkow Scholarship, Program of Excellence Scholarship.
- **Coursework:** Chemical Process Sustainability, Heterogeneous Catalysis, Advance Process Design, Structure and Properties of Materials, Separations, Kinetics, Thermodynamics, Moment Heat and Mass Transfers, Fundamentals of Chemical Engineering, Physical Chemistry, Intro to Java Programming Language, Numerical Methods, Machine Learning for DSA.

PROFESSIONAL EXPERIENCE

Research Assistant | *Electrical & Computer Engineering, University of Oklahoma* **Aug 2025 – Current**

- Conduct research on Raman spectroscopy and SERS for PFAS detection, including Ag nanoparticle synthesis and substrate preparation to enhance detection sensitivity.
- Collaborate with faculty and researchers to develop Raman/SERS-based microfluidic sensors for environmental monitoring and sustainable sensing applications.

Research Intern | *Gallogly College of Engineering at the University of Oklahoma* **Jul 2024 - Dec 2024**

- Developed methodologies to assess structural strain effects on IrO2 surfaces for oxygen evolution reaction optimization, incorporating rigorous data analysis to identify key trends.
- Executed over 80 DFT simulations to evaluate strain effects on IrO2 and Pt catalysts, achieving significant performance enhancements that lowered AEM reaction overpotential from 0.38 V for IrO2 (101) and demonstrating effective process improvement.
- Presented findings at the Great Plains Catalysis Symposium (GPCS) 2024 and co-authored a peer-reviewed article in Industrial & Engineering Chemistry Research (ACS) on strain effects in IrO2 electrocatalysts, featured in the 2025 "Celebrating Undergraduate Research in Chemical Engineering" Special Issue.

Undergraduate Researcher | *Gallogly College of Engineering at the University of Oklahoma* **Oct 2022 - Apr 2023**

- Enhanced Linux proficiency by scripting in Bash and Python while processing over 200 datasets using the OU supercomputer, aligning with process optimization initiatives.
- Reviewed over 20 research papers to extract key trends in green hydrogen technology, contributing to a 10% efficiency improvement in adsorption energy analysis methods and supporting continuous improvement efforts.
- Submitted and visualized 200+ datasets for transition metal oxide complexes to determine adsorption energies of O*, OH*, and OOH* on metal surfaces, demonstrating strong data analysis capabilities and technical problem-solving.

SKILLS

- **Process & Design**: Aspen Plus, Reaction Engineering, PFD & P&ID Development, Engineering Economic Analysis.
- **Analytical & Lab:** Raman Spectroscopy, SERS, Microfluidics, Ag Nanoparticle Synthesis.
- **Programming & Data:** Python (NumPy, Pandas, Matplotlib), Bash/Linux, VBA, Jupyter Notebook.
- **Software & Professional Tools:** AutoCAD, Microsoft Excel, Microsoft Office, Lean Six Sigma Principles.

LEADERSHIP AND PROJECTS

Project Manager | *NexLuSense LLC* **Aug 2025 – Present**

Lead projects for a Norman-based startup developing intelligent sensing and mobility technologies. Coordinate cross-functional collaboration across research and engineering teams, contribute to SBIR/STTR proposal development, and support prototype advancement from academic research toward real-world applications.

OU Housing and Residence Life | *Community Assistant* **May 2022 - Feb 2025**

Managed housing operations for 200+ residents, ensuring full compliance with documentation standards, resolving 300+ inquiries per semester, and maintaining zero safety incidents in 2022–2023.

Peace Project (Women's Empowerment and Health Initiative) **May 2024 - Aug 2024**

Educated women on sexual health and menstrual hygiene through weekly lectures, distributed 600+ sustainable menstrual products, and facilitated discussions to challenge cultural taboos in rural villages of Nepal.